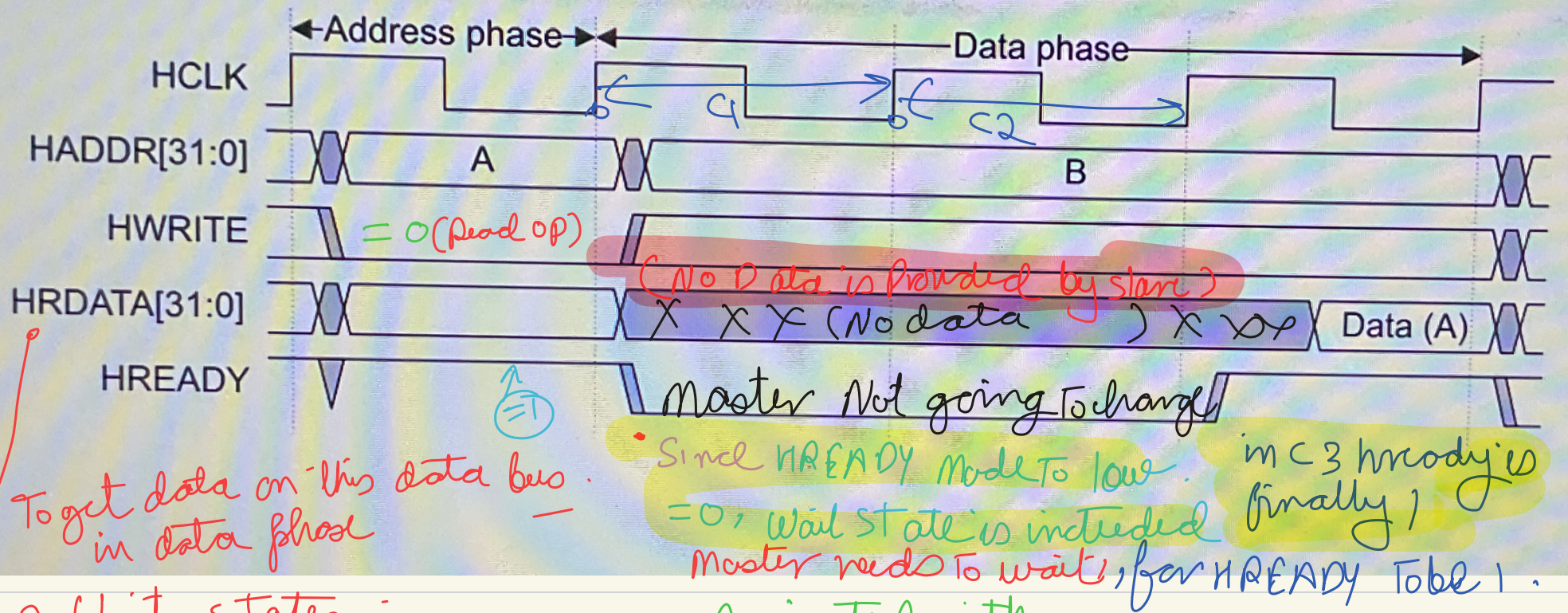


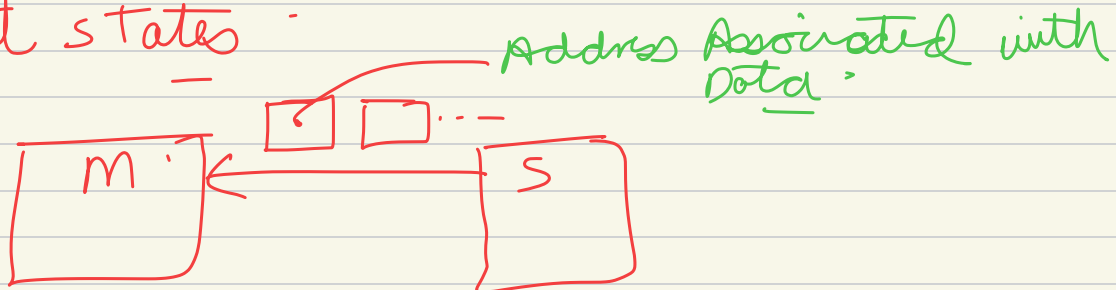




→  $HREADY = 0$ , → wait states



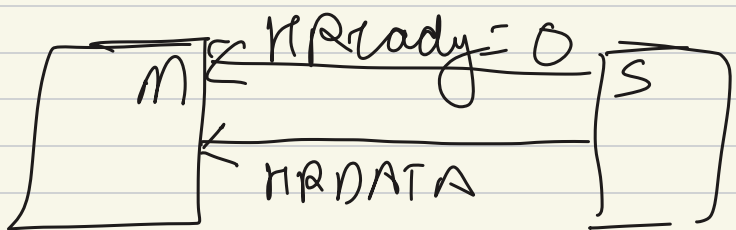
2 Wait states



Read transfer.

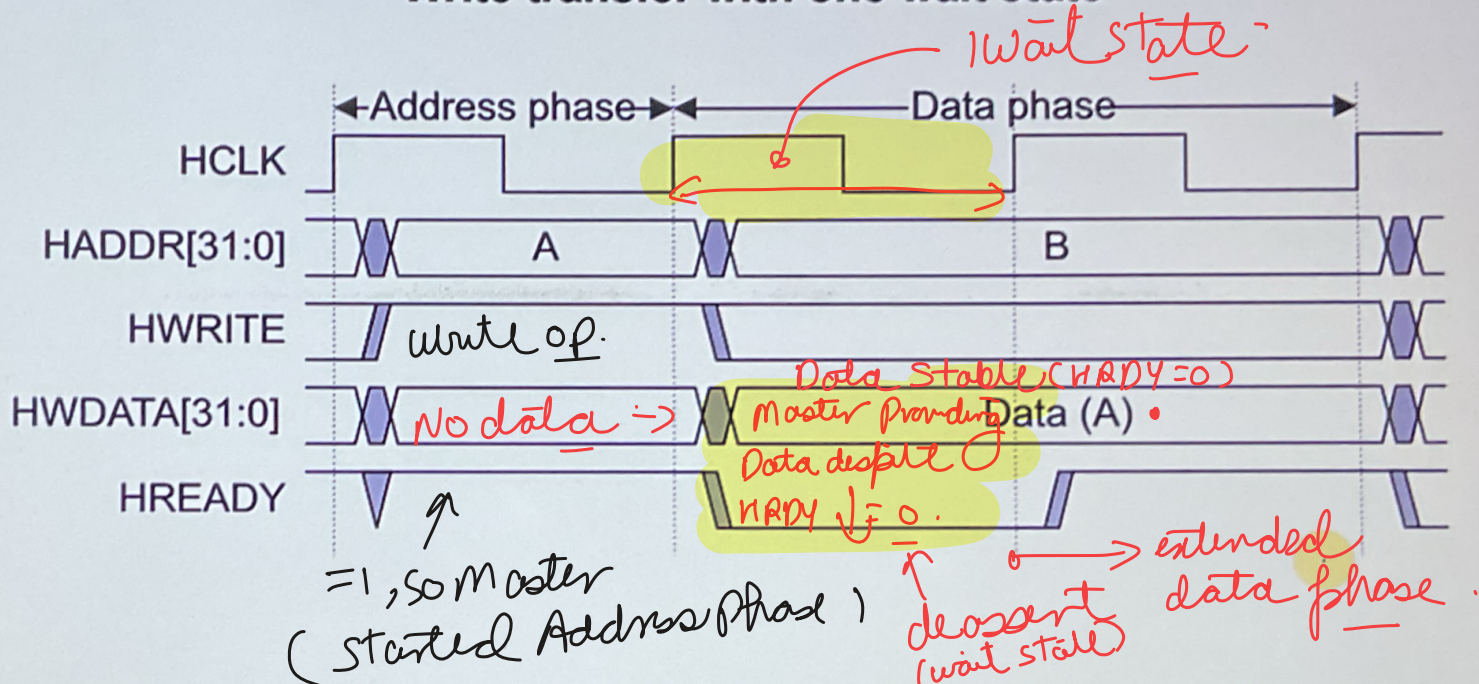
② Phases — Address (all control signals should be valid, by master)  
+ Data — write (master → slave)  
phase — read (slave → master)

→ Master won't change phase if,  $HREADY = 0$ .



$HREADY = 0$ , wait state

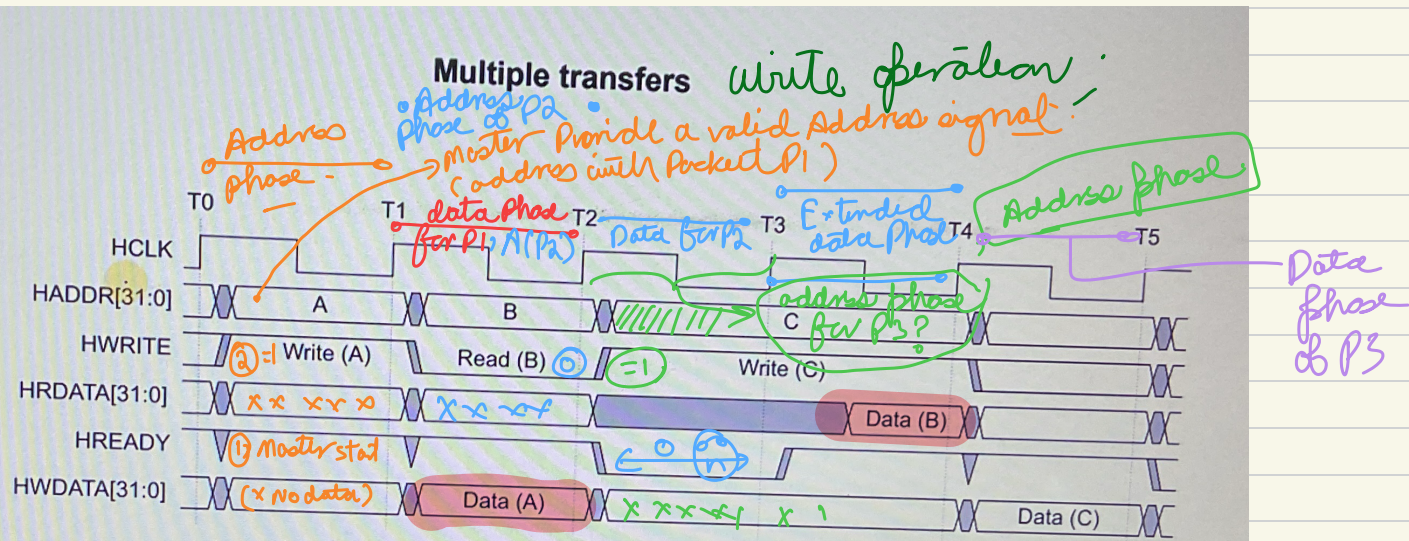
### Write transfer with one wait state



For write operations the Master holds the data stable throughout the extended cycles. For read transfers, the Subordinate does not have to provide valid data until the transfer is about to complete.

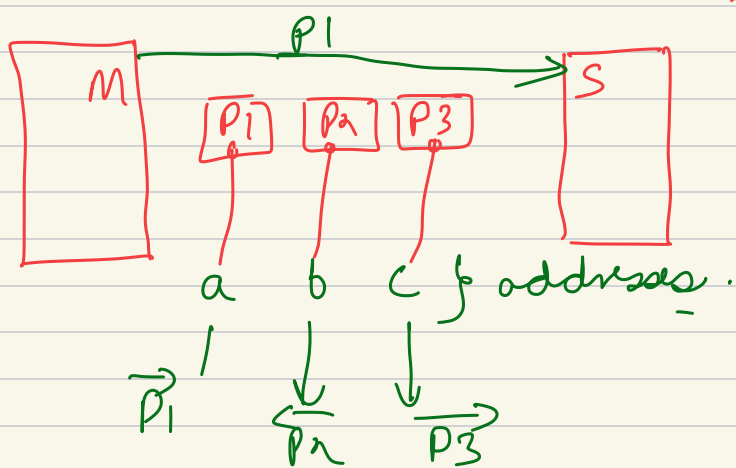


lec 4/12



Pipelined Protocol, (T<sub>1</sub> → T<sub>2</sub>) data phase for P1, for P2 (1 address + 1 data phase)

- Address phase of P2 started in data phase of packet P1



Prod) (prod) (write) How its actually happening

① check for HREADY signal.

Master won't start any phase.

③ Data phase of 1 packet can be address phase of other packet (Pipelined Protocol)